

Laws of Indices — Paper B

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper.
- The laws you need: $a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$ $(a^m)^n = a^{m \times n}$ $a^0 = 1$
- The multiplication and division laws only work when the bases are the same.
- Where a question says “as a single power”, leave your answer in index form. Where it asks for a value, work it out.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – A POWER OF A POWER

1. Write each answer as a single power.

(a) $(2^3)^2$ [1]

(b) $(5^4)^3$ [1]

(c) $(x^2)^5$ [1]

2. Simplify fully. Everything inside the bracket takes the outer power.

(a) $(3x^2)^2$ [1]

(b) $(2y^3)^4$ [1]

(c) $(5a)^3$ [1]

If your answer still has the same number in front of the letter as the question did, you have left the coefficient behind.

3. Simplify, leaving each answer as a single power.

(a) $(a^3)^4 \div a^5$ [1]

(b) $(b^2)^3 \times b^4$ [1]

4. Work out the value of each of these as an ordinary number.

(a) $(2^2)^3$ [1]

(b) $(10^2)^3$ [1]

SECTION 2 – THE ZERO INDEX

5. Complete the table. Each empty box is worth one mark. [4]

Power	Value
2^5	
5^3	
7^0	
10^4	

6. Explain why $5^0 = 1$. You must use the division law in your answer. [2]

Two marks means two statements: what the law gives you, and what the arithmetic gives you.

7. Work out the value of each of these.

(a) $2^0 + 3^0$ [1]

(b) $4^0 \times 4^3$ [1]

(c) $(6^3)^0$ [1]

SECTION 3 – PUTTING THE LAWS TOGETHER

8. Work out the value of each of these without a calculator.

(a) $2^4 \times 2^3 \div 2^5$ [1]

(b) $3^5 \div 3^3$ [1]

(c) $(10^3)^2 \div 10^4$ [1]

Simplify to a single power first. Working out $3^5 = 243$ before you divide is doing the hard version on purpose.

9. Simplify fully.

(a) $3a^2 \times 4a^5$ [1]

(b) $(2p^3)^2 \times p$ [1]

(c) $20b^8 \div 5b^2$ [1]

10. Two calculations use the same three numbers.

(a) Work out $(2^3)^2$. [1]

(b) Work out $2^3 \times 2^2$. [1]

(c) Explain why the two answers are different. [1]

11. Which is larger, 2^{10} or 10^3 ? Show your working. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.